

Hypothesis Testing

Lecture 08 · FUNDAMENTALS OF STATS AND DATA SCIENCE WITH PYTHON

From sample data to population claims

In the previous lectures, we often started from a population model or from a sampling distribution.

In practice, we usually work in the opposite direction.

- We observe a sample, compute a statistic, and ask what it says about an unknown population parameter.
- A confidence interval asks: which parameter values are plausible?
- A hypothesis test asks: is the sample evidence strong enough to challenge a benchmark claim?

From confidence intervals to tests

A confidence interval gives a range of plausible values for a parameter.

A hypothesis test takes one benchmark value and asks whether that value is difficult to reconcile with the sample.

For a simple two-sided test at the 5% level: if the 95% confidence interval excludes the benchmark, we reject it; if it includes the benchmark, we fail to reject it.

The test is not a proof. It is a decision rule under sampling uncertainty.

Hypothesis testing

In a hypothesis test, we confront two claims about an unknown population parameter.

The null hypothesis H_0 is the benchmark claim. It is the claim we test against.

- The alternative hypothesis H_1 is the competing claim. It represents the direction or type of departure we care about.

The logic is: assume the benchmark is true, measure how unusual the sample evidence would be, and reject H_0 only when the evidence is strong enough.

Example: claim = average waiting time ≤ 90 sec; $H_0: \mu \leq 90$; $H_1: \mu > 90$

Some definitions

- ❑ Null hypothesis (H_0): the benchmark claim used as the starting point of the test.
- ❑ Alternative hypothesis (H_1): the competing claim, chosen before seeing the result.
- ❑ One-sided test: the alternative is directional, for example $\mu > \mu_0$ or $\mu < \mu_0$.
- ❑ Two-sided test: any difference from the benchmark matters, for example $\mu \neq \mu_0$.
- ❑ Type I error: rejecting H_0 when H_0 is true.
- ❑ Type II error: failing to reject H_0 when H_1 is true.
- ❑ Significance level (α): the maximum Type I error probability we accept before running the test.

Example 1: one-sided marketing test

You work in a marketing firm and propose a new web campaign for a client.

The previous campaign had a click-through rate of 0.35%. You test the new campaign on a random sample of users.

The business question is directional: implement the new campaign only if there is evidence that its true CTR is higher.

Conservative default

Default: do not adopt unless evidence is higher.

H_0 : true CTR $\leq$ 0.35%

H_1 : true CTR $>$ 0.35%

Reject H_0 : adopt the campaign.

Optimistic default

Default: adopt unless evidence is worse.

H_0 : true CTR $\geq$ 0.35%

H_1 : true CTR $<$ 0.35%

Reject H_0 : do not adopt.

Example 2: two-sided quality-control test

Your company produces titanium rings that must measure exactly 5 cm in diameter.

A new production process could create rings that are too large or too small. Both deviations matter.

The benchmark is exact calibration at 5 cm.

$$H_0: \mu = 5 \text{ cm}$$

$$H_1: \mu \neq 5 \text{ cm}$$

This is a two-sided test because departures in both directions are relevant.

Choosing the significance level (α)

Before looking at the test result, decide how much evidence is required before rejecting H_0 .

The significance level α is the maximum probability we accept of rejecting a true null hypothesis.

Typical choices:

$\alpha = 5\%$: standard benchmark

$\alpha = 1\%$: stronger evidence required

$\alpha = 10\%$: weaker evidence required

Smaller α means fewer false alarms but a harder test. Larger α makes rejection easier but increases the risk of false alarms.

Procedure

- Step 1. State H_0 and H_1 , including whether the test is one-sided or two-sided.
- Step 2. Choose the significance level α before looking at the result.
- Step 3. Check the conditions for the test: randomness, independence, distributional assumptions, and sample size.
- Step 4. Compute the test statistic, or equivalently the p-value.
- Step 5. Compare with the decision rule and write a careful conclusion in words.

Evidence relative to noise

A sample estimate can differ from the benchmark just by chance.

The core comparison is:

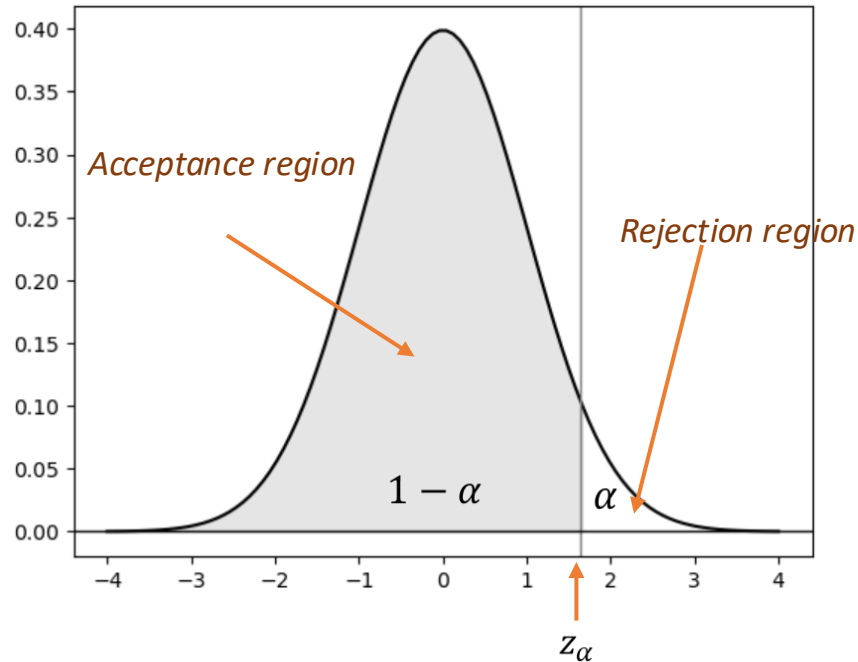
$$\text{test statistic} = (\text{estimate} - \text{benchmark}) / \text{standard error}$$

This measures the distance between the sample estimate and the benchmark in standard-error units.

A result in the desired direction is not enough. It must be large relative to sampling variability.

Critical value: right-tail test

Under H_0 , use the standard normal distribution to mark the rejection region on the right.



For any α between 0 and 1 there is a value z_α called the **critical value** that separates the normal curve into two parts. One with area $1 - \alpha$, called the *rejection region*, and the other with area α , called *acceptance region*

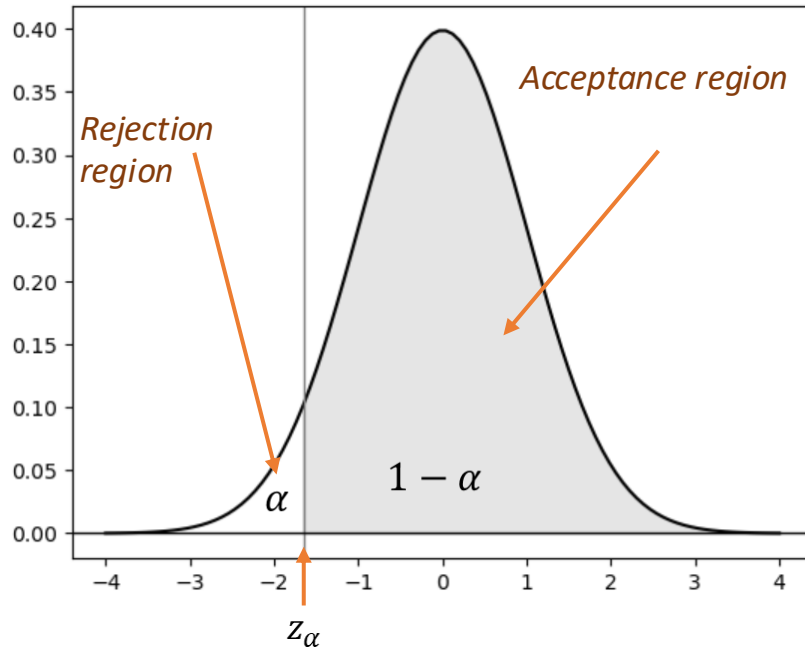
Python computes the right-tail critical value like this.

```
import scipy.stats
alpha=.05
z = scipy.stats.norm.ppf(1-alpha)
print(z)
```

1.6448536269514722

Critical value: left-tail test

Under H_0 , use the standard normal distribution to mark the rejection region on the left.

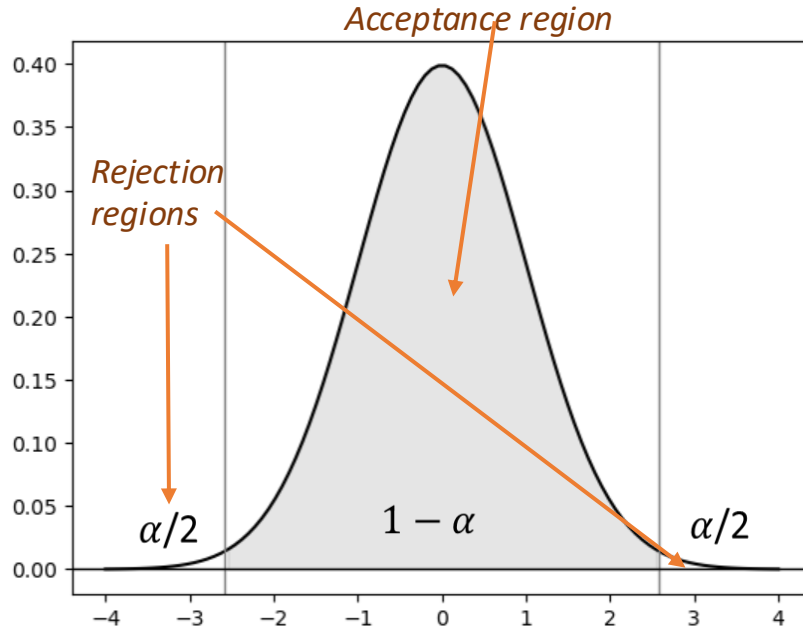


For a left-tail test, the critical value is read directly from the lower tail.

```
import scipy.stats
alpha=.05
z = scipy.stats.norm.ppf(alpha)
print(z)
```

-1.6448536269514729

Critical values: two-sided test



For a two-sided test, split α between the two tails. The rejection region is therefore in both extremes of the distribution.

```
import scipy.stats
alpha=.05
z_right = scipy.stats.norm.ppf(1-alpha/2)
z_left = scipy.stats.norm.ppf(alpha/2)
print(z_left, z_right)
```

-1.9599639845400545 1.959963984540054

P-values

A p-value answers this question: If H_0 were true, how surprising would evidence this extreme be?

- ❑ Small p-value: the sample would be unusual under H_0 , so evidence against H_0 is stronger.
- ❑ Large p-value: the sample is not very unusual under H_0 , so evidence against H_0 is weak.

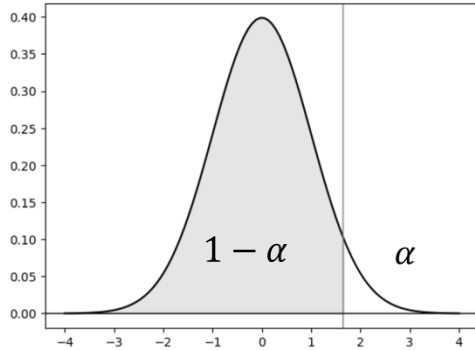
A p-value is not the probability that H_0 is true.

From evidence to a test statistic

- After collecting the sample, we convert the evidence into a standardized statistic.
- **test statistic = (estimate - benchmark) / standard error**
- For a mean, the standard error depends on how variability is handled.
- Known sigma: z-statistic, with $SE = \sigma / \sqrt{n}$
- Unknown sigma: use s ; in small samples, use a t-statistic
- In this course, we mainly use large-sample z-tests, while stating the variance assumption.

One-sided right-tail test with known variance

One-sided test



$H_0: \mu \leq \mu_0$

$H_1: \mu > \mu_0$

α is chosen by us

Known σ , and

sample size n

1. Compute the sample mean, $\bar{X}$
2. Compute the critical value z_α
3. Compute the Z-value:

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

4. Reject if $Z > z_\alpha$

- Z measures how different is the sample mean from the hypothesized population mean
- z_α represents our tolerance limit. If the measured sample is more different than this, we reject.

Writing a Statistical Conclusion

Decision language:

- If $p \leq \alpha$, reject H_0 ;
- if $p > \alpha$, fail to reject H_0 .

Failing to reject H_0 does not prove the benchmark is true. It means the sample did not provide strong enough evidence against it.

Statistical significance is not the same as practical importance.

A good conclusion reports the statistical decision, the business meaning, and the main limitations.